Postdoctoral Researcher
University of Oklahoma
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Jess Birky

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Research Interests: I aim to develop novel statistical tools for constraining and interpreting theoretical models in the the context of large-scale stellar surveys.

Education

PhD Astronomy (Data Science Program), University of Washington – Seattle	2021 — 2026
MS Astronomy, University of Washington – Seattle	2019 — 2021
BS Physics, University of California – San Diego	2015 — 2019

Professional Appointments

Postdoctoral Researcher – University of Oklahoma, Norman OK — Jul 2026 – Present
Advisor: Sean Matt

Graduate Student Researcher – University of Washington, Seattle WA — Aug 2019 – Jun 2026
Advisor: Rory Barnes, Co-advisor: James Davenport

Undergraduate Researcher – University of California San Diego, La Jolla CA — May 2016 – May 2019
Advisor: Adam Burgasser

Research Intern – Max Planck Institute für Astronomie, Heidelberg DE — Summer 2017 & 2018
Advisor: David Hogg

Selected Awards & Honors

2025	Wallerstein Dissertation Fellowship ($22k)
2019 – 2024	**NSF Graduate Research Fellowship** ($144k)
2024	UW Astronomy Excellence in Teaching Award
2024	Washington Space Grant Fellowship ($50k)
2022, 2024	UW Astronomy Jacobsen Award ($1.5k)
2018, 2019	UCSD Provost Honors (for academic achievement)
2017, 2018	Max Planck Institute Fellowship (€ 6k)
2017	UCSD Frances Hellman Research Scholarship ($5k)
2016 – 2018	UCSD Physics Chair Challenge Award ($900)

Publications

Metrics: h-index: 7 | first author citations: 120 | total citations: 269 [Google Scholar] [ADS]

Lead-author Papers (underlined = undergraduates mentored by me)

- **Birky, J.**, et al. 2026, *Physically Motivated Gaussian Processes for Stellar Variability II: Inferring Spot Temperatures from Multichromatic Lightcurves* [In prep.] [code]
- **Birky, J.**, Agol, E., et al. 2026, *Physically Motivated Gaussian Processes for Stellar Variability I: A Starspot Kernel with Differential Rotation* [In prep.] [code]
- **Birky, J.**, Barnes, R. K., 2026, *ALABI: Active Learning for Accelerated Bayesian Inference* [arxiv:2603.18259] [code]
- **Birky, J.**, Barnes, R. K., Davenport, J. R. A., 2025, *Prospects of Constraining Equilibrium Tides in Low-Mass Binary Stars*, ApJ, 992, 133 [doi10.3847/1538-4357/adf4cf] [arxiv:2507.12639]
- Hobson-Ritz, M., **Birky, J.**, Peterson, L., Gwartney, P., Delker, J., Wong, R., Gordon, T., Davenport, J. R. A., Barnes, R. K., 2024, *Tidal Synchronization of TESS Eclipsing Binaries*, ApJ, 990, 124 [doi:10.3847/1538-4357/adf10d] [arxiv:2501.04082]

- **Birky, J.**, Barnes, R. K., Fleming, D. P., 2021, *Improved Constraints for Trappist-1 XUV Luminosity Evolution*, RNAAS, 5, 122 [doi:10.3847/2515-5172/ac034c] [arxiv:2105.12562] [code]
- **Birky, J.**, Hogg, D. W., Mann, A., Burgasser, A. J., 2020, *Temperatures and Metallicities for M dwarfs in the APOGEE Survey*, ApJ, 892, 1 [doi:10.3847/1538-4357/ab7004] [arxiv:2001.04962] [code]

Co-author Papers:

- Amaral, L., Shkolnik, E., L., (incl. **Birky, J.**) 2026, *The Impact of Stellar Flares on the Atmospheric Escape of M Dwarf Exoplanets II: How TRAPPIST-1 planets lost their primordial atmospheres* [In prep.]
- Barnes, R. K., Amaral, L., **Birky, J.**, et al. 2026 *The Cummulative XUV Flux of GJ 1132* [Submitted PSJ]
- Gilbert-Janizek, S. R., et al. (incl. **Birky, J.**) 2026, *A validated whole-planet model of the Earth without life for terrestrial exoplanet studies*, Submitted PSJ [arxiv:2602.02267]
- Davenport, J. R. A., Charbour Barra, F., et al. (incl. **Birky, J.**) 2024, *Automated Spectroscopic Wavelength Calibration using Dynamic Time Warping* [arxiv:2508.05862]
- Barnes, R. K., Amaral, L., **Birky J.**, et al. 2023, *History and Habitability of the LP 890-9 Planetary System*, *PSJ*, 6, 25, [doi:10.3847/PSJ/ad94dc] [arxiv:2412.02743]
- Gialluca, M. T., Barnes, R. K., et al. (incl. **Birky, J.**) 2024, *Bayesian Calculations of Water Inventories and Oxygen Accumulation on the TRAPPIST-1 Planets*, PSJ, 5, 137 [doi:10.3847/PSJ/ad4454]
- Hsu, C., Burgasser, A. J., et al. (incl. **Birky, J.**) 2024, *The Brown Dwarf Kinematics Project (BDKP). VI: Ultracool Dwarf Radial and Rotational Velocity Survey with SDSS/APOGEE High-Resolution Spectrometer*, *ApJS*, 274, 40 [doi:10.3847/1538-4365/ad6b27] [arxiv:2403.13760]
- Hsu, C., Burgasser, A. J., et al. (incl. **Birky, J.**) 2021, *The Brown Dwarf Kinematics Project (BDKP). V. Radial and Rotational Velocities of T Dwarfs From Keck/NIRSPEC High-Resolution Spectroscopy* [doi:10.3847/1538-4365/ac1c7d] [arxiv:2107.01222] [code]
- Davenport, J. R. A., Windemuth, D., et al. (incl. **Birky, J.**) 2021, *The Rise and Fall of the Eclipsing Binary, HS Hydra*, *ApJL*, 162, 189 [doi:10.3847/1538-3881/ac1f97] [arxiv:2107.10954]
- Martin, D. V., El-Badry, K., et al. (incl. **Birky, J.**) 2021, *TOI-1259Ab–a gas giant with 2.6% deep transits and a bound white dwarf companion*, *MNRAS*, 507, 3 [doi:10.1093/mnras/stab2129] [arxiv:2101.02707]

Software Development

Lead Developer

`spotgp`: *A Python package in* `JAX` *for modeling properties of stellar starspots from lightcurves using a physically-motivated Gaussian Process.*

`parallax-presentations`: *Presentation software for creating interactive slideshows with RevealJS.*

`alabi`: Active Learning for Accelerated Bayesian Inference – *A Python package for performing Bayesian inference and optimization with computationally expensive models using Gaussian processes and active learning.*

`vplanet_inference`: *A Python package for performing parameter sweeps, model sensitivity analysis, and MCMC inference with* `vplanet` *models.*

`tikz_editor`: *A graphical interface for creating and editing LaTeX TikZ diagrams.*

Contributing Developer

`vplanet`: *A software package in C for simulating stellar and planetary system evolution, with a focus on habitability.*

`SMART`: The Spectral Modeling Analysis and RV Tool – *A Python package for forward modeling high-resolution stellar spectra with atmospheric models.*

Talks / Posters

The Role of Tides and Differential Rotation in the Angular Momentum Evolution of Low-Mass Stars 2026
PhD Thesis defense, University of Washington

Prospects of Constraining Tidal Dissipation of Low-mass Stars 2024
Poster presentation at Cool Stars 22, UC San Diego

The Formation of Short-period Binaries in Hierarchical Triples 2024
General Exam, University of Washington

Prospects of Constraining Tidal Dissipation of Low-mass Stars 2023
Qualifying Exam, University of Washington

Prospects of Constraining Tidal Dissipation of Low-mass Stars 2023
Talk at DDA Meeting 54, Lansing, Michigan

Challenges in Establishing an Accurate Model of Tidal Dissipation for Low-mass Binary Stars. 2023
Poster presentation at AAS 241, Seattle WA

(Invited) *Precise abundances of M dwarfs: data driven models applied to large scale surveys* 2022
Talk at Cool Stars 21, Toulouse, France

Constraining the XUV Luminosity Evolution of Low Mass Stars. 2022
Poster presentation at Cool Stars 21, Toulouse France

ALABI: Active Learning for Accelerated Bayesian Inference 2021
IAU Symposium 362 – Predictive Power of Computational Astrophysics, Virtual Conference

Systematic Classification of TESS Eclipsing Binaries. 2020
Poster presentation at AAS Meeting 235, Honolulu HI [poster]

Physical Parameters for 10,000+ M dwarfs in the APOGEE Survey 2019
Talk at Sloan Digital Sky Survey Collaboration Meeting, Ensenada, Mexico

Precise Stellar Parameters for 10,000+ APOGEE M dwarfs. 2019
Poster presentation at AAS Meeting 233, Seattle WA [poster]

Data-Driven Spectral Models for APOGEE M Dwarfs. 2018
Poster presentation at AAS Meeting 231, Washington DC [poster]

Modeling Stellar Parameters for High Resolution Late-M and Early-L Dwarf SDSS/APOGEE Spectra 2017
Poster presentation at AAS Meeting 229, Grapevine TX [poster]

Data Driven Models for APOGEE M dwarfs 2017
Talk at Stars Meeting & Milky Way Meeting, MPIA, Heidelberg, Germany

Identification of H-band Absorption Lines in High Resolution APOGEE Spectra of the Lowest Mass Stars. 2016
Poster presentation at the national SACNAS Conference, Long Beach CA

Telescope Time Awarded

TESS Eclipsing Binaries in Open Clusters Survey 2022 – 2023
PI: **APO 3.5 meter** – 16 total half nights with ARCES/KOSMOS spectrographs

Training the Cannon: Calibrating APOGEE Observations of Ultracool Dwarfs 2018 – 2019
f Co-I: **NASA IRTF** – 6 nights with iShell spectrograph (PI: Adam Burgasser)

APOGEE-2 Survey of the Lowest-Mass Stars and Brown Dwarfs: Composition, Chemistry and Companions 2017 – 2018
Co-I: **APOGEE 2.5-meter** – Fibers for ancillary survey (PI: Adam Burgasser)

Observing Experience

Apache Point Observatory (APO) 3.5m
6 half nights (remote), Instruments: ARCES & KOSMOS Q1 2023
8 half nights (remote), Instruments: ARCES & KOSMOS Q4 2022
4 half nights (onsite), Instruments: ARCES, KOSMOS, ARCTIC Q3 2022
2 half nights (remote), Instruments: ARCES & KOSMOS Q2 2022
2 half nights (remote), Instruments: ARCES, TripleSpec, DIS, NICFPS, ARCTIC Q4 2020

Manastash Ridge Observatory (MRO) 30in
1 full night (onsite) Q4 2023

Research Mentorship

Co-founder of YVC-UW Partnership for Research in Astrophysics (YUPRA) Summer 2024

Co-founded a partnership program between Yakima Valley Community College (YVC) and University of Washington (UW) designed to expose underrepresented students from eastern Washington to research. Received $50k grant from Washington Space grant to operate the summer 2024 REU and provide student stipends. Designed student research projects and served as primary research mentor for six YVC undergraduates in summer 2024. Project webpage: https://jessicabirky.com/yupra/

Mentor for Pre–Major in Astronomy Program (Pre-MAP) Fall 2022

Served as mentor for Pre-MAP: a program designed to expose incoming UW students to programming and/or scientific research who are traditionally underrepresented in astronomy, such as low-income and/or first-generation college students.

Students mentored:

Marshall Hobson-Ritz (UW post-bac; now PhD student at University of Maryland) Jan 2023 – Aug 2025
Karime Estrada (YUPRA REU) Summer 2024
Josue Cruz (YUPRA REU) Summer 2024
Alexandra Sanchez (YUPRA REU) Summer 2024
Michelle Marquez (YUPRA REU) Summer 2024
Julizza Gomez (YUPRA REU) Summer 2024
John Delker (UW undergrad, PreMAP program; now PhD student at Michigan State) Nov 2022 – Dec 2022
Leah Peterson (UW undergrad, PreMAP program) Nov 2022 – Sept 2023
Peter Gwartney (UW undergrad; now PhD student University of Alabama) Jun 2021 – Sept 2022
Rachel Wong (UW undergrad, co-mentored with PhD student Tyler Gordon) Jun 2021 – Sept 2022

Leadership and Service

Department Grad Student Representative 2024 – 2025

Represent and advocate for graduate student interests at department faculty meetings, manage the assignment of grad jobs, and plan graduate student orientation and community events. Also initiated the development of a grad wiki website for documenting resources.

Lead Teaching Assistant 2024 – 2025

Developed a new training program for all new TAs in the department. This includes writing a TA handbook, helping TAs develop discussion materials, providing teaching feedback through in-class observation sessions, as well as assisting in solving any class issues.

Teaching Experience

Teaching Assistant: Lead weekly lab or discussion sections, held office hours, and graded assignments/exams for both intro and advanced astronomy courses:

ASTR 421: Stellar Theory and Observations (Instructor: Emily Levesque) Winter 2025

Upper-division major course. Observations and theory of the atmospheres, chemical composition, internal structure, energy sources, and evolutionary history of stars.

ASTR 101: Introduction to Astronomy (Instructor: Chris Laws) Spring 2024

Intro course for non-science majors. Introduction to the universe, with emphasis on conceptual, as contrasted with mathematical, comprehension. Modern theories, observations; ideas concerning nature, evolution of galaxies; quasars, stars, black holes, planets, solar system.

ASTR 421: Stellar Theory and Observations (Instructor: Emily Levesque) Winter 2024

ASTR 150: The Planets (Instructor: Toby Smith) Fall 2023
Intro course for non-science majors. Survey of the planets of the solar system, with emphases on recent space exploration of the planets and on the comparative evolution of the Earth and the other planets.

ASTR 101: Introduction to Astronomy (Instructor: Chris Laws) Summer 2023

ASTR 150: The Planets (Instructor: Nicole Kelly) Spring 2021

ASTR 150: The Planets (Instructors: Nicole Kelly, Eric Agol) Winter 2021

ASTR 102: Introduction to Astronomy (Instructor: Scott Anderson) Fall 2020

Guest Lecturer: ASTR 511 (graduate galactic dynamics), ASTR 425 (undergrad cosmology)

Other Service and Outreach

Journal Reviewer, MNRAS 2025 – Present
Astronomy Grad Congress Representative Fall 2024 – Summer 2025
Planetarium Volunteer – outreach shows at the UW planetarium Fall 2023 – Present
Astronomy on Tap Seattle – Organizer & Livestream Manager Fall 2022 – Fall 2024
`VPLanet` Workshop III – SOC and Session Lead Sept 2023
Yakima Valley CC – organized overnight field trip for students to observe at MRO Oct 2023
Apache Point Observatory – Telescope Allocation Committee Fall 2020 – Fall 2022
AAS Foundations of astronomical data science workshop – volunteer helper Jan 2023
Yakima Valley CC – Outreach talk: Research in modern Astronomy Nov 2022
UW Pre-Map program – Research Mentor Fall 2022
`VPLanet` Workshop II – SOC and Session Lead Sept 2022

Engineering Experience

Design Team – UCSD Human Powered Submarine Club Sept 2015 — May 2017
Prototyped 3D submarine hull profiles to minimize fluid drag, while subject to constraints of internal diving/safety equipment. Designed submarine propulsion fin mechanism, CADed Solidworks models, and prototyped using 3D printing.

Professional Development

NBIA Summer School on Astrophysical Dynamics of Gravitating Systems – *Copenhagen, Denmark* Aug 2024
AI-driven discovery in physics and astrophysics – *Tokyo, Japan* Jan 2024
Cool Stars 20.5 – *Virtual Conference* Mar 2021
NExSS Quantitative Habitability Science Workshop – *Online workshop* Dec 2020
online.tess.science – *Online workshop* Sep 2020
TESS Ninja 3: Expanding the Science of TESS – *Sydney, Australia* Feb 2020
ZTF Collaboration Meeting – *UW Seattle, WA* Sept 2019
Caltech FUTURE of Physics Workshop – *Pasadena, CA* Nov 2018
M33 HST Survey Meeting – *Ringberg Castle, Tegernsee, Germany* Jul 2018
Conference for Undergraduate Women in Physics – *Cal Poly Pomona, CA* Jan 2018
Gaia Sprint – *Internationales Wissenschaftsforum Heidelberg, Germany* Jul 2017
Conference for Undergraduate Women in Physics – *UC Los Angeles, CA* Jan 2017

References

Prof. Rory Barnes (UW) – PhD advisor `rkb9@uw.edu`
Prof. James Davenport (UW/DIRAC) – PhD co-advisor `jrad@uw.edu`
Prof. Adam Burgasser (UCSD) – undergrad research advisor `aburgasser@ucsd.edu`
Prof. David Hogg (NYU/MPIA/Flatiron) – undergrad research advisor `david.hogg@nyu.edu`